

Do Kieu Trang Thoi

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EDUCATION

Virginia Tech – United States <i>PhD in Industrial and Systems Engineering</i>	2024 – 2028 GPA: 3.79/4.00
Ewha Womans University – South Korea <i>Master's in Electronics and Electrical Engineering</i>	2022 – 2024 GPA: 4.11/4.30
Ho Chi Minh City University of Technology – VNU, Viet Nam <i>Bachelor's in Industrial Systems Engineering</i>	2017 – 2022 GPA: 8.27/10.00
University of Limerick – Ireland <i>Erasmus Student Mobility</i>	2019 (Fall) GPA: 3.65/4.00

WORK EXPERIENCE

Virginia Tech, Grado Department of Industrial & Systems Engineering <i>Graduate Research Assistant (GRA) & Graduate Teaching Assistant (GTA)</i>	<i>Blacksburg, U.S</i> 08/2024 – Present
– Conduct research on disease prediction using large language models (LLMs) applied to medical claim reports, electronic health records (EHR) datasets and collaborate with data scientists from Carilion Clinic. – Develop and analyze statistical models to improve efficiency, human–AI collaboration. – Teaching Assistant for ISE 4404 Statistical Quality Control (Fall 2025), supporting lectures, grading assignments, and assisting students with statistical process control methods. – Lab Instructor for ISE 3424 Discrete-Event Simulation (Spring 2026), instructing 2 lab sections (30 junior ISE students each) and guiding students in simulation modeling using Risk and Simio. – Main Instructor for ISE 3424 Discrete-Event Simulation (Summer 2026).	
Intelligent Sensing & Processing Laboratory <i>Student Researcher / Teaching Assistant</i>	<i>Seoul, Korea</i> 06/2022 – 07/2024
– Researched deep learning for image processing, ultrasound reconstruction, and robot-assisted ultrasound (UR3e). – Supported teaching activities and supervised student projects.	
Nestlé Vietnam <i>Engineering Intern</i>	<i>Dong Nai, Vietnam</i> 05/2021 – 12/2021
– Led a Maintenance Cost dashboard (Power BI, VBA). – Built Shift Handover, Checklist, and Tool Storage apps (Power Apps, Power Automate, SAP scripting). – Contributed to an automated production scheduling project (regional scope).	
The Dow Chemical Company <i>E-Project Participant</i>	<i>Ho Chi Minh City, Vietnam</i> 03/2020 – 07/2020
– Project: Application of digitalization in Warehouse Management System. – Performed data analysis, designed database/UI, and assisted software development.	

- Applied text mining to identify contractual risk clauses in EPC documents.
- Developed Python scripts for keyword extraction and clause mapping.

AWARDS AND HONORS

- Alexander Walter Scholarship (Spring 2026) 01/2026
- Virginia Tech Graduate Research Assistant Fellowship (Fall 2024, Spring 2025) 08/2024
- Brain Korea 21 (BK21) Scholarship 09/2023
- Ewha Global Partnership Program (EGPP) — Full-ride scholarship 08/2022
- Academic Scholarship for Top 5% Excellent Students 2018 & 2021
- AmCham Women in Engineering Scholarship (ACWES) 11/2020
- Gold Medal, Design–Manufacturing–Application Award 11/2020
- Consolation Prize, “Students with Start-up Ideas (SV.STARTUP)” 12/2020
- Most Effective Solution, eProjects Innovation Showcase 07/2020
- Erasmus+ Mobility Scholarship by European Commission 09/2019
- Student of 5 Merits Award 09/2018

PUBLICATIONS

1. **Thoi, D.K.T.**, McNamara, R.S., Savla, J., Zhong, H., 2026. *Hybrid Sampling with Structured-Text Features for Chronic Pain Prediction in Cognitive Impairment*. Proceedings of the **International Conference on AI in Healthcare 2026**.
2. **Thoi, D.K.T.**, McNamara, R.S., Savla, J., Zhong, H., 2026. *A Semi-Supervised Multi-Modal Framework for Early Cognitive Impairment Detection in Chronic Pain Patients*. Manuscript under submission.
3. **Thoi, D.K.T.**, Tran, H., Thoi, D.N.T., Zhong, H., 2026. *The Enhanced Physics-Informed Kolmogorov–Arnold Networks: Applications of Newtons Laws in Financial Deep Reinforcement Learning (RL) Algorithms*. [\[paper\]](#)
4. ***Thoi, D.K.T.**, Seo, K., Lim, J.H., Park, J.S., Yang, M.J., Song, T.J., Park, S., 2025. *Diagnosis of invasive pancreatic cancer in endoscopic ultrasound images leveraging translation models*. **Biomedical Signal Processing and Control** 102, 107389. [\[paper\]](#)
5. **Thoi, D.K.T.**, Lim, J.H., Park, J.S., Park, S., 2025. *Deep learning-assisted diagnosis of extrahepatic common bile duct obstruction using MRCP imaging and clinical parameters*. [\[paper\]](#)
6. **Thoi, D.K.T.**, Seo, K.W., Park, S., 2024. *Deep Supervised Domain Adaptation for Invasive Pancreatic Cancer Diagnosis from Endoscopic Ultrasound Images*. **Proc. SPIE** 12932, Medical Imaging 2024: Ultrasonic Imaging and Tomography. [\[paper\]](#)
7. Pham, T.T.A., **Thoi, D.K.T.**, Choi, H., Park, S., 2023. *Defect Detection in Printed Circuit Boards Using Semi-Supervised Learning*. **Sensors**, 23(6), 3246. [\[paper\]](#)

SKILLS

Programming Languages	Python, MATLAB, R, Docker
ML & DL Frameworks	scikit-learn, Keras, PyTorch, TensorFlow
Simulation & Design Tools	AutoCAD, Arena
Database & Statistical Software	SQL, Power BI, VBA, Minitab, SPSS, JMP
Languages	Vietnamese (Native), English (IELTS 6.5 – CEFR B2)

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- Vice President, INFORMS Student Chapter at Virginia Tech, United States (08/2025 – 07/2026)
- President, IE Connect Club, Ho Chi Minh City University of Technology (06/2018 – 07/2019)
- Fully Funded Delegate, CommTECH Insight Camp, Surabaya, Indonesia (01/2019 – 02/2019)

REFERENCES

Huaiyang Zhong, Ph.D.

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Suhyun Park, Ph.D

Professor, Department of Electronic & Electrical Engineering, Ewha Womans University
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Nguyen Van Phuc Nguyen, Ph.D.

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